Adhvaith Vijay (U.S. Citizen)

Advancing Automated Defect Detection at Tesla

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WORK EXPERIENCE

Tesla | Senior Data Scientist **July 2024 - Present**

- Qualify the next generation of Tesla battery cells by forecasting risk of thermal runaway over battery warranty life with telemetry data from 3.5 million vehicles
- Develop and deploy computer vision algorithms (U-Net architecture) to segment 50+ battery components from CT scans, delivering 98% recall on critical checks across 2,000+ newly manufactured batteries daily - every newly manufactured TESLA 4680 battery passes through my algorithms as the final quality gate before entering Tesla vehicles.
- Built a Celery/Redis ETL pipeline on 6 VMs that runs CT inference in under 1 min/scan across 2,000+ batteries/day, logs metrology (e.g. electrode overhang, core concavity, cathode gap) to a central database, and triggers Splunk alerts on model-flagged out-of-spec cells; feeds real-time quality dashboards, saving 200+ technician hours/week

Standard AI | Data Scientist **Dec 2023 - June 2024**

- Built a marketing measurement solution using real-time 3D pose data to analyze churn, customer cohorts, and interaction-to-purchase conversions, generating $600K in new revenue for client in 3 months.
- Led in-store A/B tests at Compass Group USA to identify the most effective product placements near store exits, increasing interaction-to-purchase conversion by 7 percentage-points
- Developed a Monte Carlo simulation engine to model shelf-to-shelf transitions and guide product layout decisions, driving changes that lifted entrance-area revenue by 15%

Tesla | Cell Engineer Intern **Jan 2023 - Sept 2023**

- Developed a Vision Transformer-based model to detect core impingement and collapse in battery cells from CT scans, improving defect detection recall by 40%
- Eliminated the need for multiple $27,000 Keyence machines by creating a CT scan–based method to non-invasively measure terminal/lid bulge on cells, achieving accuracy within 50 microns
- Built a Tableau dashboard, auto-updated daily, to analyze battery cell failure patterns by integrating dissection report data with production metrics, enabling identification of defect trends and improving quality control processes

JPMorgan Chase & Co. | Summer Associate (Credit Fraud Risk) **June 2022 - Aug 2022**

- Improved PySpark-based small business loan default model by 14% within the Consumer & Community Banking team by engineering domain-specific features that captured erratic repayment patterns, seasonal revenue shifts, and spikes in credit utilization
- Enhanced model interpretability for my team's lending systems by applying feature binning and hierarchical clustering on model input features, enabling clear explanations of high-risk applicant segments for federal regulatory audits

Rolls-Royce North America | Undergraduate Researcher **Oct 2019 - May 2020**

- Created an algorithm to determine the optimal compressor wash interval for Rolls-Royce AE3007 engines, enhancing maintenance efficiency
- Deployed my algorithm as a R Shiny application that reduced the time to identify flights needing compressor washes from 3-4 hours to under 5 minutes, enabling geospatial visualization and isolation of hundreds of at-risk aircraft

EDUCATION

University of California, San Diego **Sept 2022 - March 2024**
Master of Science in Data Science

University of California, Los Angeles **Sept 2019 - June 2022**
Bachelor of Science, Statistics and Data Science

SKILLS

Languages & ML: Python, R, SQL, scikit-learn, PyTorch, XGBoost
Data & ML Ops Tools: Airflow, dbt, Git, AWS, Databricks, Docker, Snowflake, H20
Analytics & Visualization: A/B Testing, Tableau, Power BI, Looker, Plotly
Domain Expertise: Lithium-Ion Battery Reliability, Shopper Behavior Analytics, Consumer Banking Risk

PATENTS

Morales Saiki, L. Y., Vijay, A., Zanlungo, F., Dhingra, A., & Lee, M. J. (2024). *Subject-tracking in a cashier-less shopping store for autonomous checkout for improving item and shelf placement and for performing spatial analytics using spatial data and the subject-tracking* (U.S. Patent Application No. US2024193497A1)

PAPERS

A. M. Vijay, F. Zanlungo, D. M. Woollard, and L. Y. Morales. Analyzing the shopping journey: Computing shelf visits in a physical retail store. In Proceedings of the 34th IEEE International Conference on Robot and Human Interactive Communication (RO-MAN), Under review.